

Maintain Control

What a control system has to contain before it can hold anything steady, and what each part looks like when it fails. Silverthorn chapter 1, plus the homeostatic reflex pathways section of chapter 6.

ALL 12 WEEK 1 COMPETENCIES

CH 1 AND THE REFLEX HALF OF CH 6

Your pre-work, in order

Three steps. Do them in this order and you finish the week with one complete sheet in your own handwriting, which is the thing that makes the exam easier.

1a

Print the Competency Study Guide One box per competency. The boxes are the competencies, so this is also your list of what you have to be able to do.

1b

First pass: this page and the book

YOU ARE HERE Fill in what you can from these notes and from Silverthorn, before you watch anything. Leave the gaps you cannot fill. The gaps are the point.

1c

Second pass: watch and add Watch the week's lectures and add what they give you that the reading did not. Use a different color so you can see which half came from where.

What physiology is

Physiology

The study of the normal functioning of a living organism and its parts, including the physical and chemical processes involved.

Anatomy

The study of structure. Physiology asks what that structure does and by what mechanism.

Pathophysiology

What function looks like when a normal mechanism fails. Not a separate subject, the same mechanisms running wrong.

Emergent property

A property of the whole system that none of its individual parts possesses. Blood pressure, consciousness, urine concentrating ability.

Teleological answer

An answer that says what something is for. Useful shorthand, not an explanation.

Mechanistic answer

An answer that says how something happens. This is what physiology means by an explanation, and what earns credit.

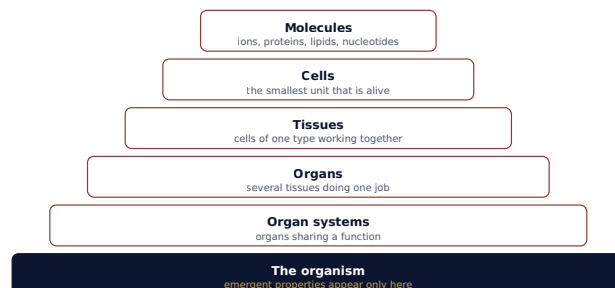


Figure 1. Six levels of organization. Function at one level cannot be fully predicted from the level below it, which is why molecules alone do not explain a heartbeat.

The internal environment

Homeostasis has to be defending something specific. This section says what it is, and the boxes show what sits inside what.

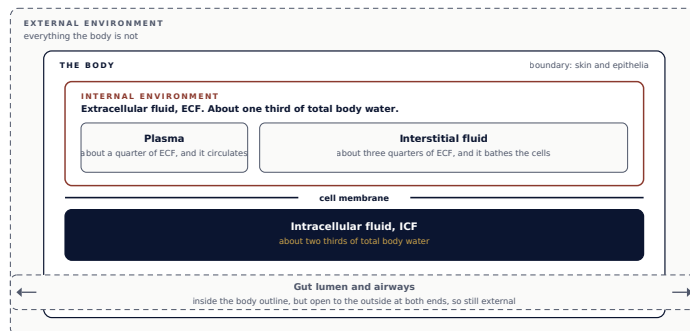


Figure 2. Read it from the outside in. The gut and the airways are drawn crossing the body wall on purpose: food sitting in your stomach has not entered you yet.

Outside the body dashed gray

Internal environment, the ECF red outline

Inside the cells, the ICF solid navy

OUTSIDE THE BODY

Everything in this box is external, including one thing that feels internal.

External environment

Everything the body is not. Air, water, food, temperature, whatever you are standing in.

Gut lumen and airways

Open to the outside at both ends, so still external. A substance is only inside you once it has crossed an epithelium.

INSIDE THE BODY, OUTSIDE THE CELLS: THE INTERNAL ENVIRONMENT

This is what homeostasis defends. When a lab value is out of range, this is the fluid it is out of range in.

Internal environment

The extracellular fluid. Claude Bernard called it the milieu interieur. Cells never touch the outside world, only this.

Extracellular fluid, ECF

All fluid outside cells. Roughly one third of total body water, split into interstitial fluid and plasma.

Interstitial fluid

ECF sitting between cells. Roughly three quarters of the ECF, and the fluid that actually bathes most tissue.

Plasma

The liquid part of blood, roughly a quarter of the ECF. The only ECF compartment that circulates quickly, which is why it is the one you sample.

INSIDE THE CELLS

Separated from the internal environment by one cell membrane, and held at a completely different composition.

Intracellular fluid, ICF

Fluid inside cells. Roughly two thirds of total body water, so the larger of the two compartments.

Why the split matters

Sodium is high outside and low inside. Potassium is the reverse. Those two gradients power most of what you study this term.

NUMBERS WORTH CARRYING

Total body water is about **60 percent** of body mass in an average adult male, closer to 50 percent in an average adult female.

Of that, roughly **two thirds is ICF** and **one third is ECF**.

Of the ECF, roughly **three quarters is interstitial fluid** and **one quarter is plasma**.

NURSING

Where the extra fluid is sitting is what changes the things you see when you assess the patient.

Example: A patient can be visibly swollen, with fluid sitting in the interstitial space, and still be low on volume inside the vessels.

RADIOLOGIC TECHNOLOGY

Contrast goes into the blood first and then leaks out into the tissue. So when you scan decides which of those two you are actually looking at.

Example: An arterial-phase scan catches contrast still inside the vessels. A delayed scan catches it after it has moved out into the interstitial space. Same injection, two different-looking images, because the contrast moved compartments.

MEDICINE

Before you pick an IV fluid you have to decide which fluid space you are trying to fill.

Example: A liter of isotonic saline stays largely in the extracellular fluid. A liter of 5 percent dextrose in water spreads across total body water, so much less of it stays in the vessels.

RESPIRATORY THERAPY

Mucus in the airway counts as outside the body, in the same way that food sitting in your gut is still outside you.

Example: A dry airway is a local humidity problem. It is not the same thing as whole-body dehydration, and it is not fixed the same way.

Homeostasis is not equilibrium

Homeostasis

The maintenance of a relatively stable internal environment. The word relatively is load bearing: the variable oscillates, it does not sit still.

Dynamic steady state

Material and energy move continuously, but the amount in each compartment stays roughly constant. This is what homeostasis actually is.

Equilibrium

A state in which no further net change occurs and no energy is required to maintain it. A cell at equilibrium with its surroundings is dead.

Regulated variable

The quantity the body is actually holding in range, such as core temperature, plasma pH, or blood glucose.

Set point

The target value the integrating center compares against. Set points move: they reset in fever, drift on a circadian cycle, and shift with acclimatization.

Normal range

The band of values a variable is allowed to occupy. Usually defined to cover the middle 95 percent of a healthy population, so one healthy person in twenty falls outside it.

Equilibrium	Steady state
Costs no energy to hold	Costs ATP every second
Concentrations end up equal	Gradients are kept unequal
Nothing is being defended	Actively defended against change
A dead cell reaches it within hours	Only a living cell can hold it

Figure 3. The sodium gradient is the standard example. Extracellular sodium near 145 mM, intracellular near 12 mM, held apart by a pump spending ATP every second of your life.

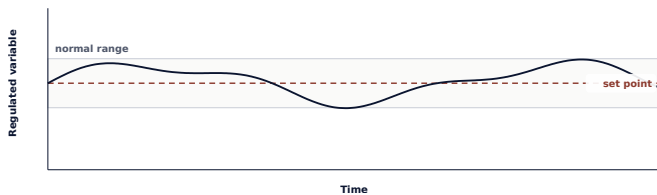


Figure 4. The variable is essentially never exactly at the set point. Negative feedback cannot prevent a change, only correct it, so oscillation is the normal result.

NURSING

When blood flow drops, cells can no longer hold their gradients. You see that in the patient before you see it in the labs.

Example: Cool mottled skin and a rising lactate are gradient failure becoming visible at the bedside.

RADIOLOGIC TECHNOLOGY

A nuclear medicine tracer only builds up where cells are still alive and still spending energy. The picture is a map of which tissue is still working.

Example: Technetium sestamibi concentrates in heart muscle because living mitochondria maintain their charge gradient. Dead tissue cannot, so it takes up nothing. That is what makes the cold spot on the scan.

MEDICINE

Every gradient in the body costs energy to hold, and that bill never stops while the patient is alive.

Example: When perfusion stops, the sodium potassium pump stops with it, potassium leaks out of cells, and measured plasma potassium rises.

RESPIRATORY THERAPY

Breathing is muscle work, and that work is the price of keeping carbon dioxide moving out.

Example: A patient who is tiring can no longer afford that gradient, which is why carbon dioxide starts to climb as they wear out.

The six themes

Every mechanism this term sits on at least two of these. Naming which two is a fast way to check that you understood a mechanism rather than memorized it.

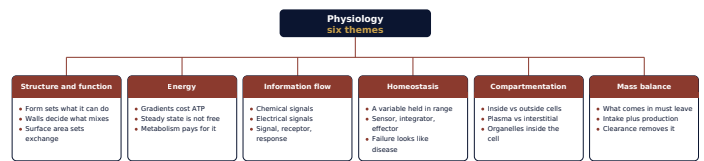


Figure 5. Use this as the spine of your Week 1 Competency Study Guide. One box per branch.

The quantitative tools

Total body load = intake + production – excretion – metabolism

Load is the total amount of a substance in the body. Not a concentration.

What it says: **a concentration can change for two entirely different reasons.** Either the amount of substance changed, or the volume it is dissolved in changed. Never conclude the first without ruling out the second. A rising plasma sodium can mean sodium was gained or water was lost, and the treatments are opposite.

Mass flow = concentration × volume flow

Units: mg/min = mg/mL × mL/min.

What it says: **delivery is not the same as concentration.** A normal oxygen concentration with a collapsed cardiac output still starves the tissue. This equation is why shock kills people whose blood looks normal.

Clearance = rate of removal ÷ plasma concentration

Units: mL/min. Clearance is a volume per unit time, not an amount per unit time.

What it says: **clearance is the volume of plasma completely cleared of a substance per minute**. No real volume is ever fully cleared. Expressing removal this way makes it independent of how much of the substance happens to be present, which is why it is the number used to set drug doses.

NURSING

Intake and output charting is just what went in minus what came out. Record it loosely and the problem hides in the gap.

Example: A patient in positive balance across three shifts has gained fluid, and the daily weight usually shows it before anything else does.

RADIOLOGIC TECHNOLOGY

Contrast follows the same rule as everything else. What you put in stays in the patient until the kidney clears it out.

Example: A patient who had contrast yesterday has not finished clearing it today. A repeat study too soon adds to a load that is still there, which is why prior studies get checked before the next injection.

MEDICINE

A number can rise because there is more of the substance, or because there is less water. Those two need opposite treatments.

Example: A rising sodium can mean sodium was gained or water was lost. Treating the wrong one moves the patient in the wrong direction.

RESPIRATORY THERAPY

Extra fluid ends up in the lung, and water in the lung gets in the way of gas crossing over.

Example: A patient several liters positive becomes harder to oxygenate, and the chest film tends to follow the balance sheet.

WORKED EXAMPLE

Plasma glucose is 90 mg/dL and renal plasma flow is 600 mL/min. Glucose delivered to the kidney per minute:

Convert: 90 mg/dL = 0.9 mg/mL.

Mass flow = 0.9 mg/mL × 600 mL/min = **540 mg/min**.

Essentially none of it appears in urine, so something is moving 540 mg per minute back across an epithelium

against its gradient. That transporter is the subject of Week 14.

Control systems

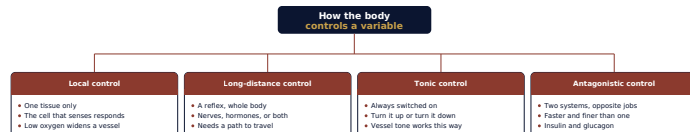


Figure 6. Local control needs no signaling system at all, which makes it fast and limited. Reflex control needs a path, which makes it slower and general.

Local control

A response restricted to the tissue where the change was detected. The cell that senses the change also produces the response. Example: falling tissue oxygen dilates nearby arterioles.

Long-distance control

A reflex involving the nervous system, the endocrine system, or both, so a change in one place produces a response somewhere else.

Tonic control

Signal is continuously on and the response is graded by changing its rate. Sympathetic tone in blood vessels works this way.

Antagonistic control

Two systems push the variable in opposite directions. Insulin and glucagon on blood glucose. Faster and finer than a single system could be.

NURSING

Knowing whether the response is happening in one spot or across the whole body tells you what to expect when you do something about it.

Example: Warming one limb changes blood flow in that limb without changing the patient's core temperature.

RADIOLOGIC TECHNOLOGY

Contrast only lights up tissue that is actually getting blood. So the bright areas are a picture of which local vessels are open.

Example: A tumor lights up because its own vessels are dilated and leaky, a purely local change. Nothing whole-body happened. The bright spot is local blood flow made visible.

MEDICINE

Local control does not need nerves or hormones, so it keeps working even when the long distance systems have failed.

Example: In a transplanted heart the nerves are cut, and local metabolic control of the coronary vessels still works normally.

RESPIRATORY THERAPY

The lung sends blood to the parts that are getting air, and it does that locally, with no instruction from the brain.

Example: Low oxygen in one region narrows the vessels there, pushing blood toward regions that are actually being ventilated.

USE THE SEVEN BOXES AS A DIFFERENTIAL

When a regulated variable is out of range, the fault is in one of the boxes. Break the sensor and nothing is detected. Break the afferent path and nothing arrives. Break the integrating center and the comparison itself is wrong. Break the efferent path or the effector and the command is issued but nothing happens. This list is the first thing to write when your patient file goes wrong.

NURSING

Name which part of the loop is broken and you know which sign to watch for, and which sign is never going to show up.

Example: In a patient on a beta blocker, do not wait for a rising heart rate to tell you they are bleeding. Watch the other signs.

RADIOLOGIC TECHNOLOGY

Reactions on the table fall into two groups depending on which pathway is firing, and the two look nothing alike.

Example: A patient who goes pale, sweaty and slow-pulsed after an injection is a vagal reaction: the efferent nerve output dropped the heart rate. Anaphylaxis drives the heart rate up instead. Same table, opposite pathways, opposite responses from you.

MEDICINE

The seven boxes give you a list of suspects. Each broken box produces its own picture.

Example: A patient who cannot raise their heart rate while bleeding may have an intact sensor and a blocked efferent path, which is exactly what a beta blocker does.

RESPIRATORY THERAPY

Nearly everything you do starts with a chemoreceptor noticing something. That is the sensor box of the loop.

Example: If the sensor is blunted, the drive to breathe falls even though the stimulus telling it to rise is still there.

The reflex loop

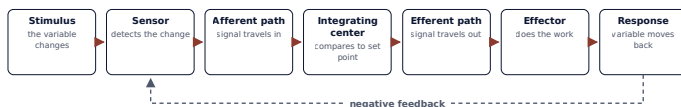


Figure 7. Seven boxes. Every reflex in this course is this diagram with different contents, so learn the shape once.

Stimulus

The change in the regulated variable that starts the loop.

Sensor or receptor

The structure that detects the change. If it fails, the body never learns anything happened.

Afferent pathway

Carries the signal toward the integrating center. Afferent means arriving.

Integrating center

Compares the incoming signal with the set point and decides on a response. Fever is a fault here, not in the sensors.

Efferent pathway

Carries the command outward. Efferent means exiting.

Effector

The muscle or gland that does the physical work of the response.

Response

The change produced, which feeds back on the sensor and closes the loop.

Response loop

The whole seven-step sequence taken together.

Feedback types

Three patterns cover almost every loop in this course. They differ in one thing only: what the response does to the stimulus that triggered it.

PATTERN	WHAT THE RESPONSE DOES	WHAT IT IS FOR	HOW IT STOPS
Negative feedback	Opposes the stimulus	Holding a variable near a set point	Shuts itself off as the variable comes back
Positive feedback	Reinforces the stimulus	Driving a process to completion	Needs an outside event to end it
Feedforward	Acts before the variable has changed at all	Blunting a change you can see coming	Ends when the anticipated load arrives or does not

Reading it this way keeps you out of the most common trap. Positive feedback is not the body malfunctioning and negative feedback is not the body being cautious. Negative means the response subtracts from the stimulus and positive means it adds to it. That is the whole distinction, and it says nothing about whether the outcome is good for the patient.

NEGATIVE FEEDBACK

The response **opposes** the stimulus that produced it. Stabilizing. It shuts itself off as the variable returns toward the set point. It cannot prevent the initial change, only correct it, which is why regulated variables oscillate.

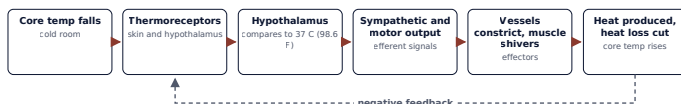


Figure 8. Thermoregulation worked all the way through. Reproduce this from memory with the real anatomy in each box.

NURSING

A body that is still fixing the problem looks different from one that has run out of room. Telling those apart is most of assessment.

Example: A brief drop on standing that recovers is the loop working. A drop that stays down is the loop failing.

RADIOLOGIC TECHNOLOGY

Every time you sit a patient up or lay them flat you are running this loop live and watching what it does.

Example: Sitting a patient up from a supine scan, or standing them for an upright film, drops their pressure for a few seconds before the reflex catches it. In an older or dehydrated patient the catch is slow, and that is the moment they go down. It is why you move them in stages and stay with them.

MEDICINE

Negative feedback cannot stop a change from happening. It can only pull it back. So the value always moves first.

Example: Blood pressure dips on standing before the reflex catches it. The dip is normal. Failing to recover from it is not.

RESPIRATORY THERAPY

Breathing harder is the body's negative feedback answer to carbon dioxide going up.

Example: Rising carbon dioxide normally drives ventilation up. If the carbon dioxide is high and ventilation is not rising, the loop is broken somewhere.

POSITIVE FEEDBACK

The response **reinforces** the stimulus, pushing the variable further from where it started.

Destabilizing by design, so it needs an event outside the loop to stop it.

Oxytocin in labor, stopped by delivery. The clotting cascade, stopped when the vessel seals. Sodium entry during an action potential, stopped by channel inactivation.

Positive feedback with no off switch describes dying: uncontrolled hemorrhage and runaway hyperkalemia both work this way.

FEEDFORWARD CONTROL

The body responds to a signal that **predicts** a change, before the regulated variable has moved.

Salivating and secreting gastric acid at the smell of food. Heart rate rising in the seconds before exercise begins.

Feedforward buys time. It cannot correct an error, because at that moment there is no error.

THE DISTINCTION, ONE LINE EACH

Negative feedback reacts to what already happened. Feedforward reacts to what is about to happen. Positive feedback commits to what is happening now.

Set points move

Fever is a deliberately raised set point, not a broken thermostat. That is why you shiver and feel cold while your temperature is already climbing: the body reads 37 C (98.6 F) as too cold against a new target.

Circadian rhythms move set points on a daily cycle. Core temperature, cortisol and blood pressure all have a normal range that depends on the hour.

Acclimatization resets over days to weeks. Hematocrit rises at altitude, sweat sodium falls in a hot climate.

NURSING

Cool a patient who is shivering and you are fighting the thermostat instead of turning it down.

Example: Shivering generates heat, so external cooling during the rise can drive the temperature higher rather than lower.

RADIOLOGIC TECHNOLOGY

Scanner rooms are kept cold for the machine. The patient's thermostat reacts to that whether you want it to or not.

Example: A cold, uncovered patient shivers, and shivering is movement, and movement is motion artifact. The warm blanket is an image quality tool, not just a comfort measure.

MEDICINE

In a fever the thermostat is not broken. It has been turned up, and the body is working hard to reach the new number.

Example: The patient shivers while their temperature is already climbing, because the body reads its current temperature as too cold against the new target.

RESPIRATORY THERAPY

If carbon dioxide stays high for long enough, the sensor stops treating it as an emergency and settles on a new normal.

Example: In long-standing carbon dioxide retention the body adapts to a higher normal, so the textbook number is no longer that patient's target.

Double blind study

Neither the participant nor the person measuring the outcome knows.

Reference range

The band of values covering the middle 95 percent of a healthy reference population. Not a definition of health.

Biological variability

Real differences between healthy people. Not measurement error, and not something to be averaged away.

WHY ONE FLAGGED VALUE IS NOT AN ANSWER

If a reference range covers the middle 95 percent of healthy people, then each test flags one healthy person in twenty. Run a twenty-analyte panel on a completely healthy person and the odds are close to even that something comes back flagged. A flagged value is a reason to look further, not an answer. You will meet this again in your patient file: every week you get one more set of numbers for the same person, and the story only shows up across all of them.

NURSING

One number on its own tells you very little. What that number did since last time tells you a lot.

Example: A creatinine at the top of normal that doubled since yesterday is a bigger problem than one that is mildly high and unchanged.

RADIOLOGIC TECHNOLOGY

You check lab numbers before giving contrast. A number outside the normal range does not automatically mean stop.

Example: A creatinine sitting slightly high and unchanged for a year is a different situation from one that doubled this week. Both print with the same flag on the report. Only one of them changes what happens next.

MEDICINE

Normal ranges are built to cover most healthy people, not every healthy person. So about one healthy person in twenty gets flagged on any given test.

Example: Run a twenty-analyte panel on a completely healthy person and something comes back flagged more often than not.

How we know any of this

Independent variable

What the experimenter deliberately changes.

Dependent variable

What is measured in response to that change.

Control

An otherwise identical condition in which the independent variable is not changed.

Placebo

An inactive treatment used to separate the effect of a substance from the effect of being treated.

Blind study

The participant does not know which group they are in.

RESPIRATORY THERAPY

A blood gas is one photo of something that is still moving.

Example: The same value on the way up and on the way down mean opposite things about where the patient is heading.

What this week assumes you already have

This is the Unit 1 prerequisite list, and it covers Week 1 and Week 1 together. If any line here is shaky, fix it before you work the week rather than after. Unit 1 has a short self-check for each of the three groups.

CHEMISTRY

- Ions and charge
- Concentration and gradients
- Water, polarity and solubility
- Acids, bases and pH
- Buffers
- Proteins and shape
- ATP and energy
- Units and simple calculation

Chemistry self-check, 10 questions

ANATOMY

- The cell and its parts
- The four tissue types

This is a physiology course, so anatomy shows up only where you need it to follow a mechanism. Two items is the whole list for Unit 1, and that is not an oversight.

Anatomy self-check, 3 questions

MATH

- Metric prefixes and unit conversion
- Counting particles, not molecules
- Rates, and what per means
- Reading a graph
- What a log scale does
- In minus out
- Charge, and why mEq is not mM
- Sanity checking an answer

Math self-check, 9 questions

Close the page and produce it

These are the 12 competencies for this week, straight off the study guide, with the two prompts that go with each one. Pick one prompt per competency. Blank paper, nothing open.

Read the full instructions first, including how the first pass before the lecture differs from the second pass after it.

01

Physiology and levels of function

Define physiology and place a given process at the correct level of organization from molecule to organism, then state the level at which that process is best explained.

Do one of the two. Head your paper with the number.

1-1

Draw a ladder with the levels of organization on it, from molecule at the bottom to whole organism at the top. Take one thing that happens in the body, write it beside the level where you would see it happening, then draw an arrow down to the level that actually explains it.

1-2

Draw one thing you have seen a person do or experience. Around your drawing, sketch what is happening at three different levels for that same event and label each level. Circle the level a physiologist would use to explain it and write one line saying why that level and not the others.

02

Structure and function relationship

Predict how a change in the structure of a molecule, cell, tissue, or organ alters its function, using examples from two different organ systems.

Do one of the two. Head your paper with the number.

2-1

Pick a structure from one organ system and draw it. Then draw it again with one thing about its structure changed. Beside the second drawing write what its function can no longer do, and why that follows from the change rather than being a separate problem.

2-2

Draw two structures from two different organ systems that are shaped the way they are for the same reason. Label the shared reason. Then for each one write what would go wrong if that feature were taken away.

03

Homeostasis defined

Define homeostasis, regulated variable, and setpoint, and distinguish homeostasis from chemical equilibrium and from steady state.

Do one of the two. Head your paper with the number.

3-1

Draw three small sketches side by side: chemical equilibrium, steady state, and homeostasis. Label what is the same in all three and what is different. Under each, write one example from the body or from ordinary life.

3-2

Draw a regulated variable being held near its setpoint over time, with the setpoint marked. Show what the line looks like when the system is working. Then write what a flat reading does and does not tell you about what is happening underneath it.

04

Feedback loop components

Diagram a negative feedback loop labeling stimulus, sensor, afferent path, integrating center, efferent path, effector, and response, and trace body temperature or blood glucose through every step.

Do one of the two. Head your paper with the number.

4-1

Draw a negative feedback loop with every part labeled: stimulus, sensor, afferent path, integrating center, efferent path, effector, response. Then run body temperature through it, writing what each part actually is in that example.

4-2

Draw the negative feedback loop again from memory, then run blood glucose through it. Now show two or three places the loop could break, and beside each one predict what you would see in the person if it broke there.

05

Negative and positive feedback

Distinguish negative from positive feedback by the direction of the response and give a physiological example of each, explaining why a positive feedback loop needs an outside event to end it.

Do one of the two. Head your paper with the number.

5-1

Draw a negative feedback loop and a positive feedback loop side by side. Mark the difference with an arrow on each. Under the positive one, write a real example from the body and then write what ends it, because a positive loop that nothing ends does not stop.

5-2

Write down three positive feedback examples from the body. For each one, draw what is amplifying and write the outside event that shuts it off. Then write one sentence on why the ending is the part people leave out.

06

Feedforward control and acclimatization

Explain anticipatory feedforward control and acclimatization and identify which one is operating in a given scenario.

Do one of the two. Head your paper with the number.

6-1

Draw a timeline of something happening to the body, and mark where a feedforward response would sit compared with where a feedback response would sit. Write what feedforward is anticipating and what it costs if the prediction is wrong.

6-2

Write two short scenarios, one where feedforward control is operating and one where acclimatization is. Draw what is happening in each and label which is which, then write the one question you would ask to tell them apart.

07

Local and reflex control pathways

Distinguish a local control pathway from a long distance reflex pathway and classify a given response as one or the other.

Do one of the two. Head your paper with the number.

7-1

Draw a local control pathway and a long distance reflex pathway on the same page. Label what does the sensing and what does the responding in each, and mark how far the signal travels.

7-2

Write down three responses the body makes. Draw each one, label it local or reflex, and write the feature of your drawing that decided it. If one is hard to classify, say what makes it hard.

08

Mass balance

Apply the mass balance equation to a solute or to body water and calculate the intake, production, and output combination that holds the amount in the body constant.

Do one of the two. Head your paper with the number.

8-1

Write the mass balance equation from memory. Then draw a person with arrows for everything coming in and everything going out for water in one day. Put numbers on your arrows that balance, and mark what would change first if one arrow got bigger.

8-2

Take a solute and draw the intake, production and output that would hold the amount in the body constant. Then change one of the three and show what has to happen to the others to keep the total steady. Write what happens if nothing else changes.

09

Units and unit conversion

Convert among the units used in physiology including molarity, osmolarity, milliequivalents, mmHg, liters per minute, and percent solutions.

Do one of the two. Head your paper with the number.

9-1

Draw a conversion map. Put each unit this course uses in its own box, then draw arrows between the ones you can convert between and write the conversion along each arrow. Mark the two you keep getting wrong.

9-2

Work three conversions you find awkward, showing every step with the units carried through. Beside each answer, draw the place in the body where you would meet that number, so the number has somewhere to live.

10

Graphing and data interpretation

Construct a labeled graph with the independent variable on the x axis, and read slope, direction, and trend from a physiological data set.

Do one of the two. Head your paper with the number.

10-1

Draw a set of axes and label them properly, with the independent variable in the right place. Sketch a physiological relationship you already know. Then mark on it where the slope is steep, where it is flat, and write what each of those means about the body.

10-2

Sketch three different graph shapes: a straight line, a curve that flattens, and a curve that rises then falls. Beside each write what that shape is telling you, and give a physiological example of each.

11

Experimental design and controls

Identify the hypothesis, independent variable, dependent variable, and control condition in a physiology experiment and state what the control rules out.

Do one of the two. Head your paper with the number.

11-1

Draw a small physiology experiment as a diagram: what you change, what you measure, and what you hold still. Label the hypothesis, the independent variable, the dependent variable and the control condition on the drawing. Then write what your control rules out, in a full sentence, because that is the part that gets marked.

11-2

Draw an experiment with something wrong in its design. Mark on the drawing what is missing or confounded. Then draw the fixed version beside it and write what the fix now lets you conclude that you could not conclude before.

12

Measurement error and variability

Distinguish random from systematic error and explain why physiological measurements are repeated and averaged.

Do one of the two. Head your paper with the number.

12-1

Draw a target with shots on it showing random error, then another showing systematic error, then another showing both. Label each. Under them write how you would detect each kind in a set of real measurements.

12-2

Draw the same measurement taken five times, plotted as five dots. Draw it again for a measurement with a systematic error in it, so the dots sit together but in the wrong place. Label which is which, and mark on each drawing whether repeating and averaging would fix it.

Dr. Sharilyn Rennie · BIO 005 Human Physiology · Maintain Control. Values given are typical teaching figures, not laboratory reference ranges.